

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

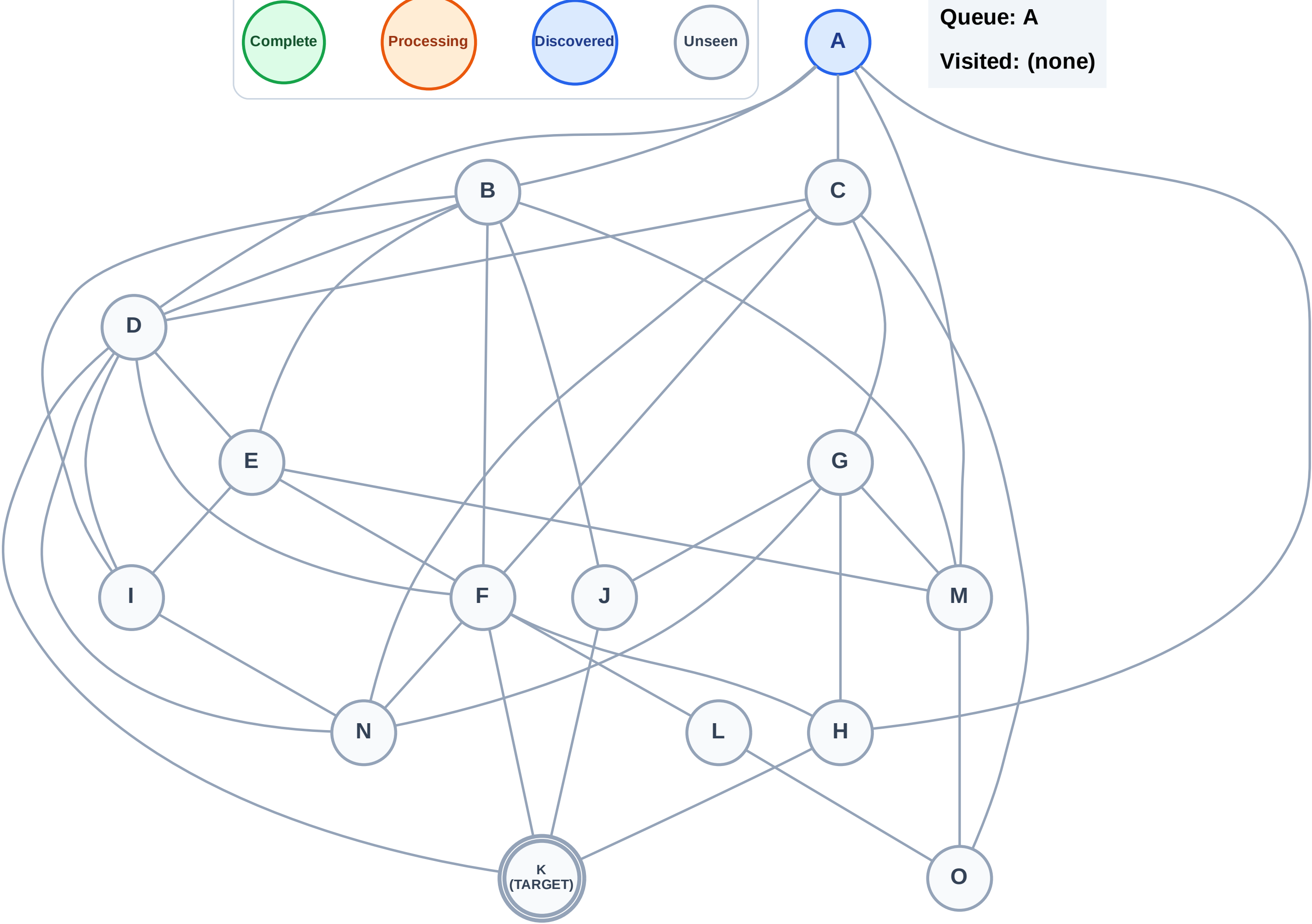
Complete

Processing

Discovered

Unseen

Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

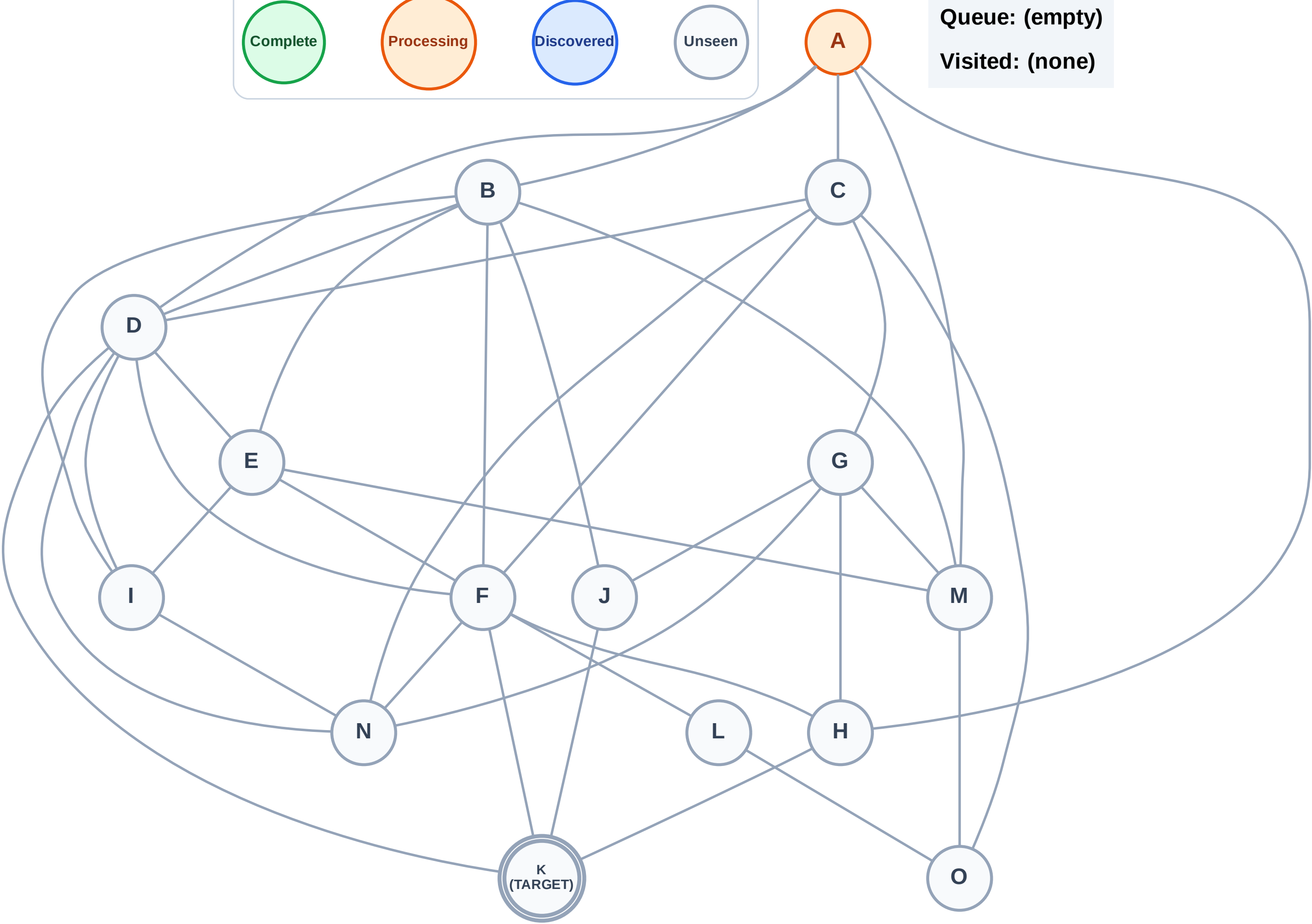
Complete

Processing

Discovered

Unseen

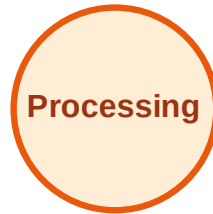
Queue: (empty)
Visited: (none)



BFS Traversal

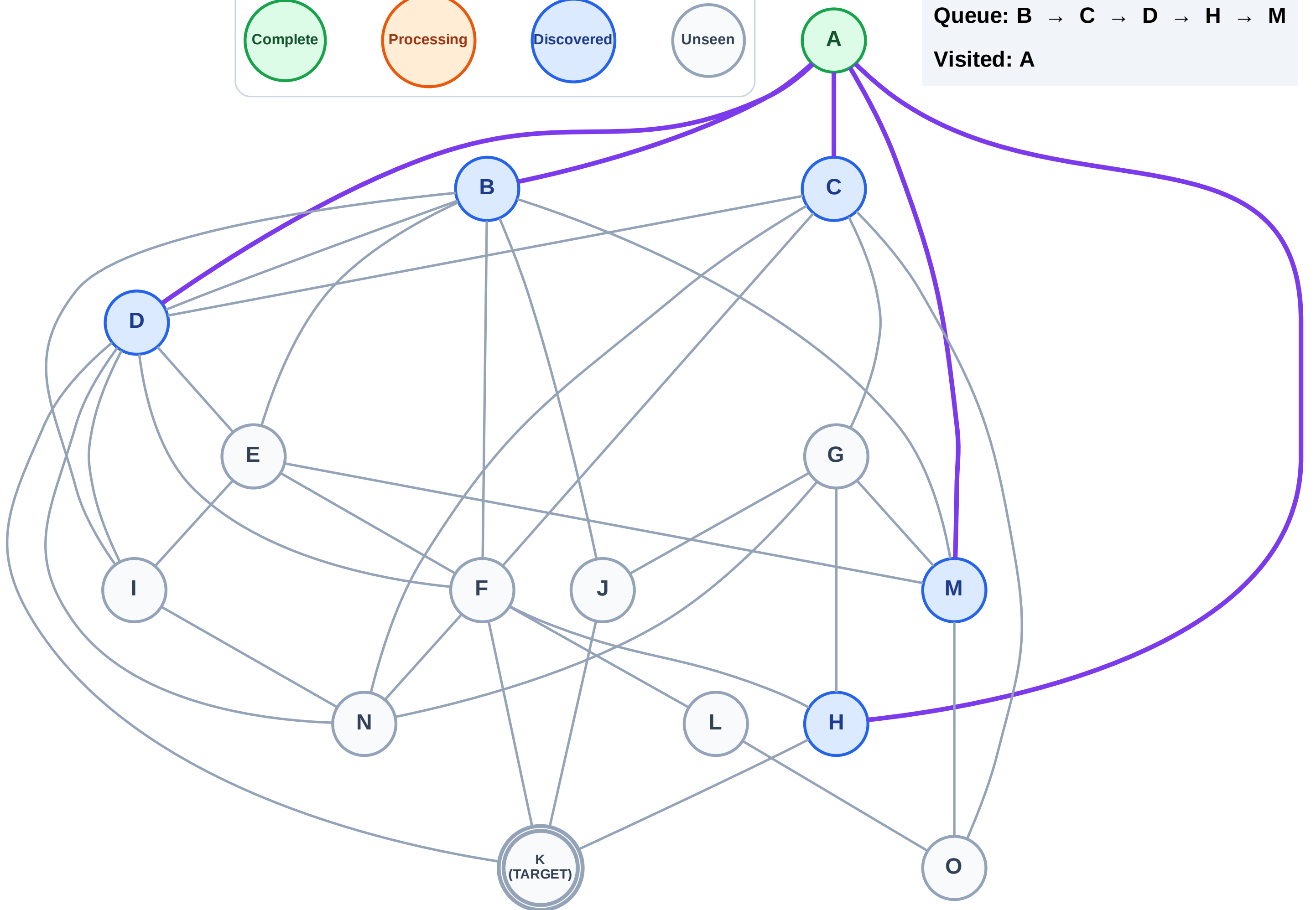
Step 3: Discover B, C, D, H, M from A

Legend



Queue: B → C → D → H → M

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

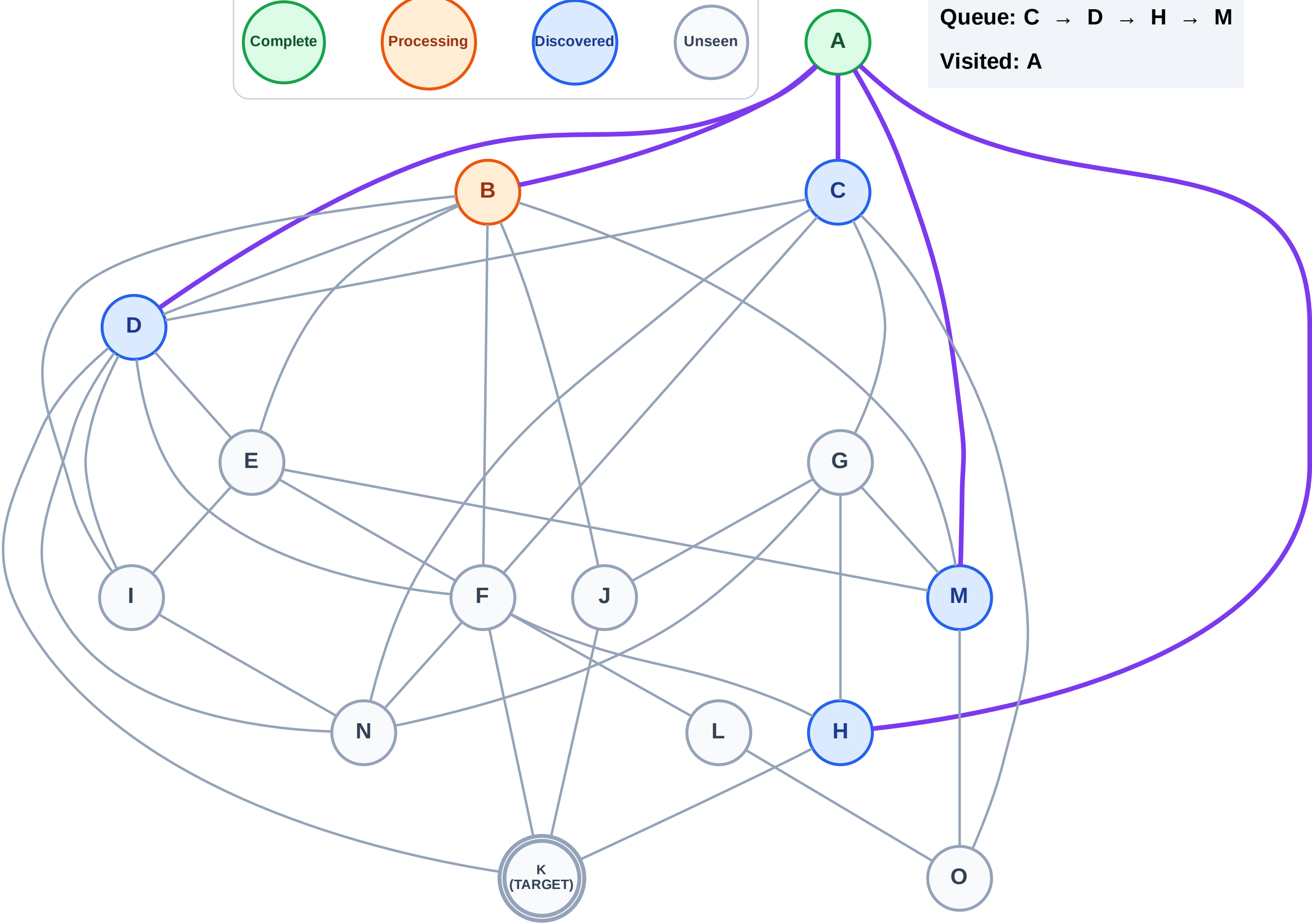
Processing

Discovered

Unseen

Queue: C → D → H → M

Visited: A



BFS Traversal

Step 5: Discover E, F, I, J from B

Legend

Complete

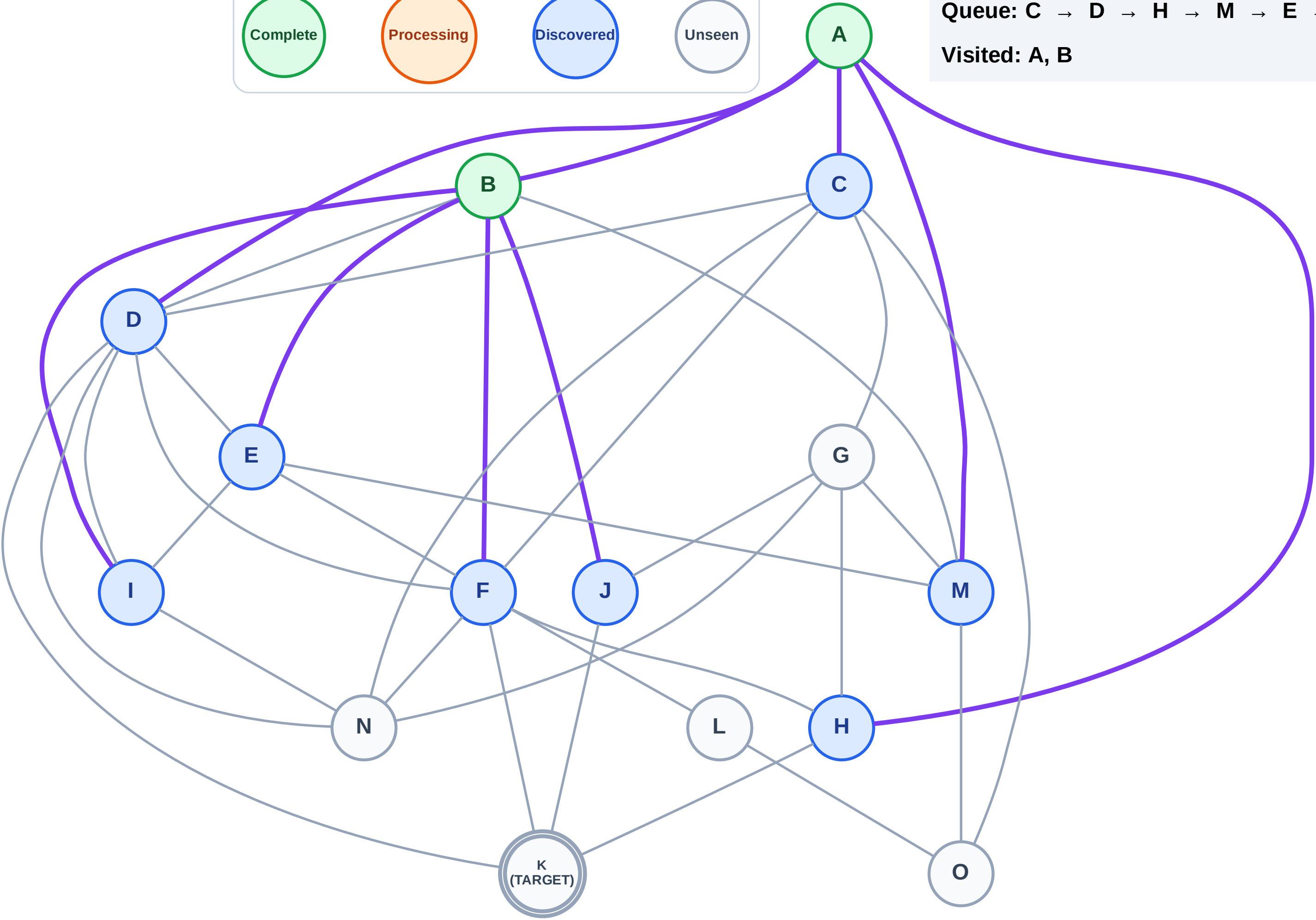
Processing

Discovered

Unseen

Queue: C → D → H → M → E → F → I → J

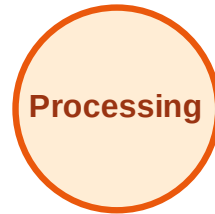
Visited: A, B



BFS Traversal

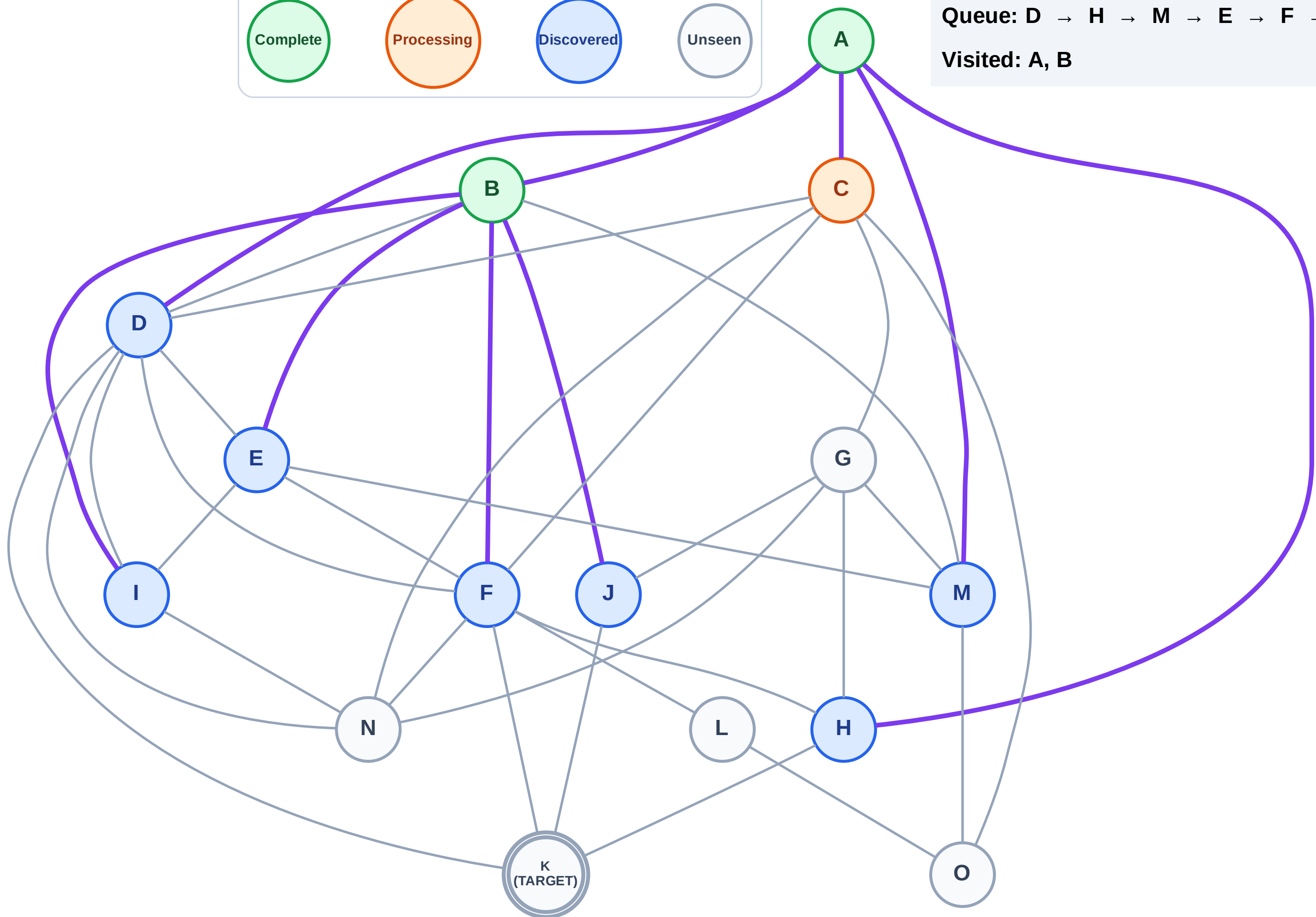
Step 6: Process node C

Legend



Queue: D → H → M → E → F → I → J

Visited: A, B



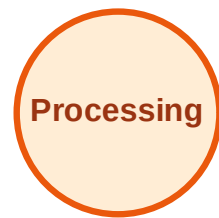
BFS Traversal

Step 7: Discover G, N, O from C

Legend



Complete



Processing



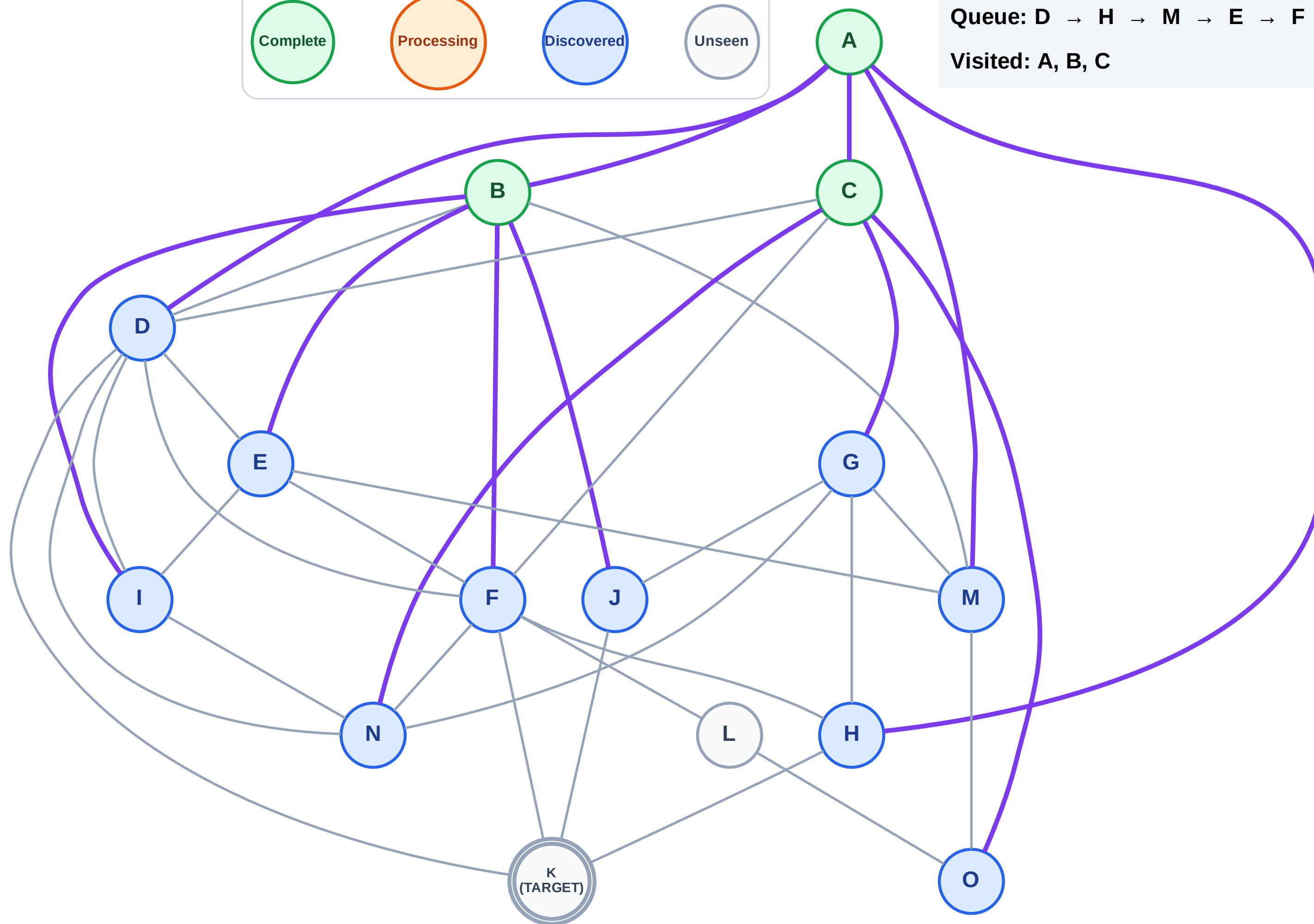
Discovered



Unseen

Queue: D → H → M → E → F → I → J → G → N → O

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

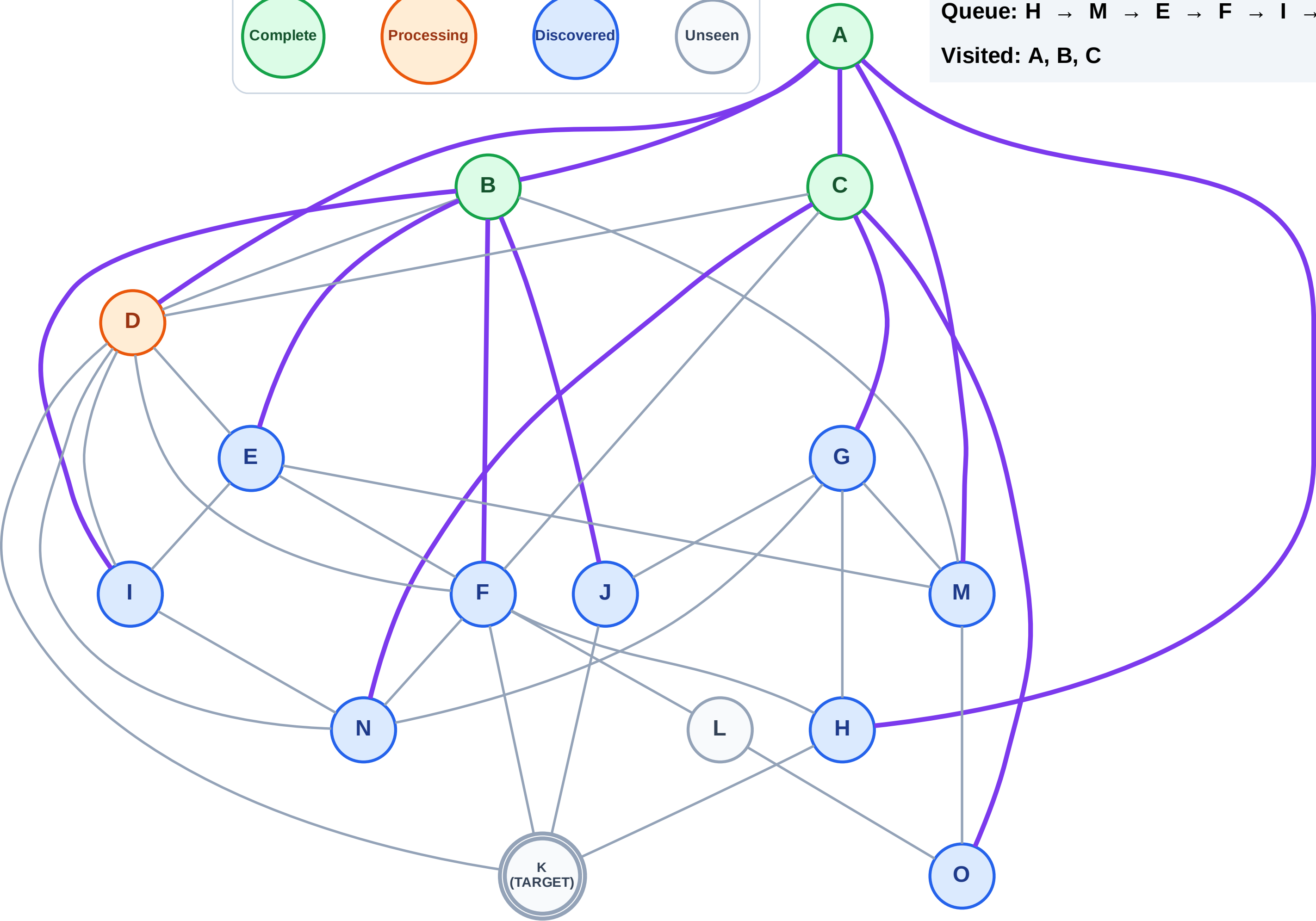
Processing

Discovered

Unseen

Queue: H → M → E → F → I → J → G → N → O

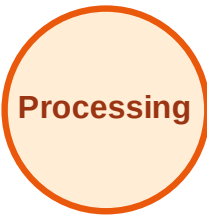
Visited: A, B, C



BFS Traversal

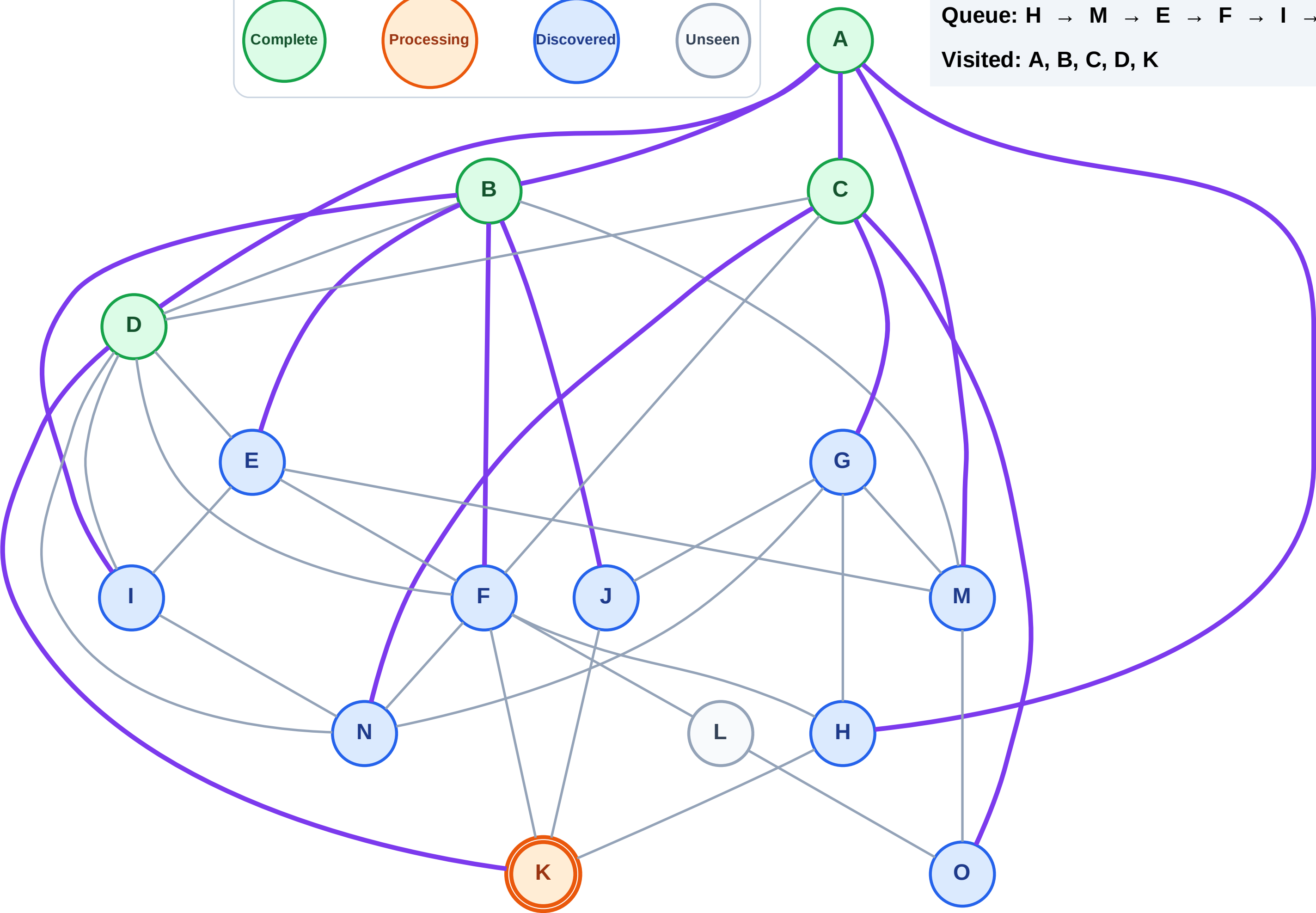
Step 9: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → D → K

Legend



Queue: H → M → E → F → I → J → G → N → O

Visited: A, B, C, D, K



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

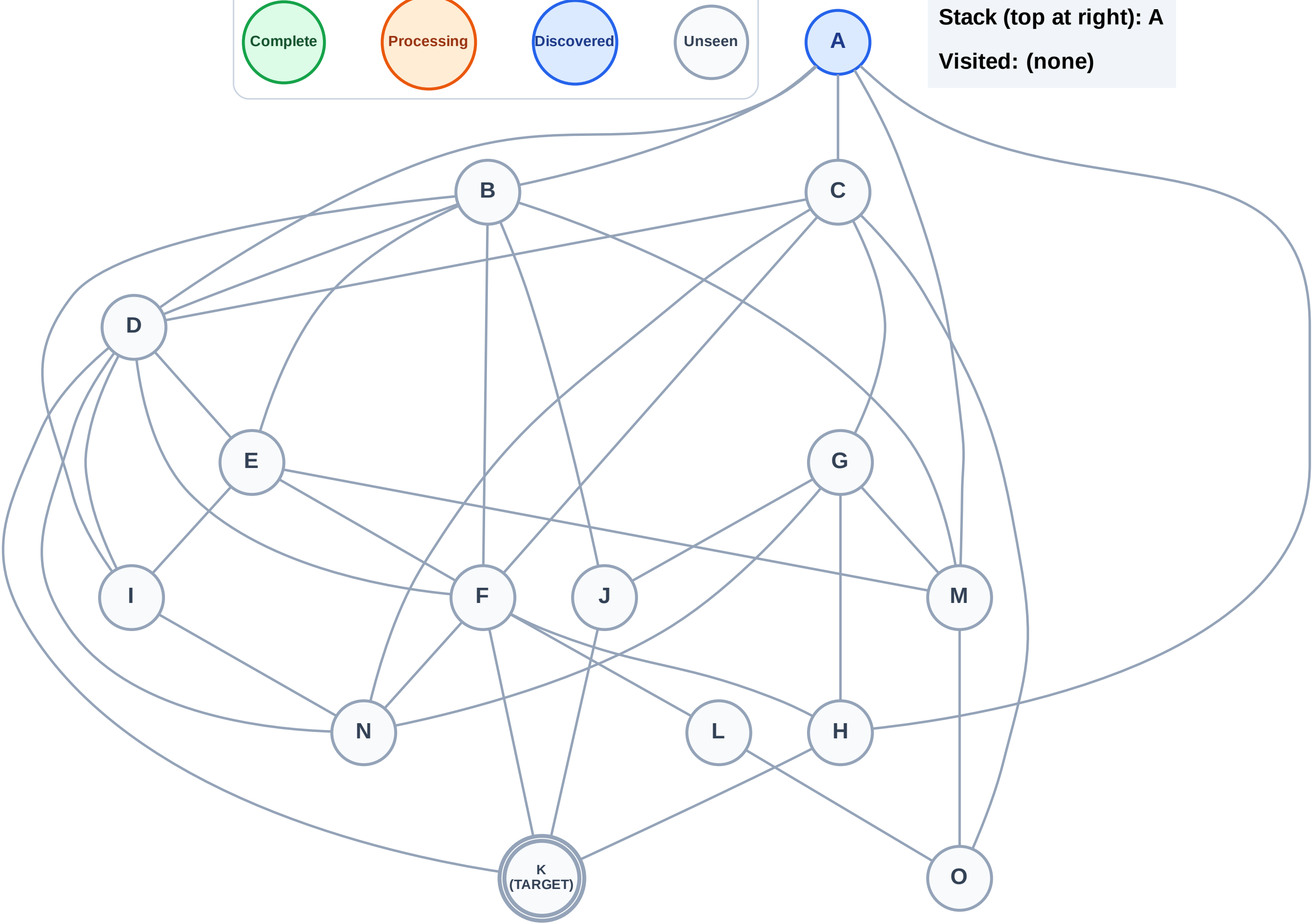
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

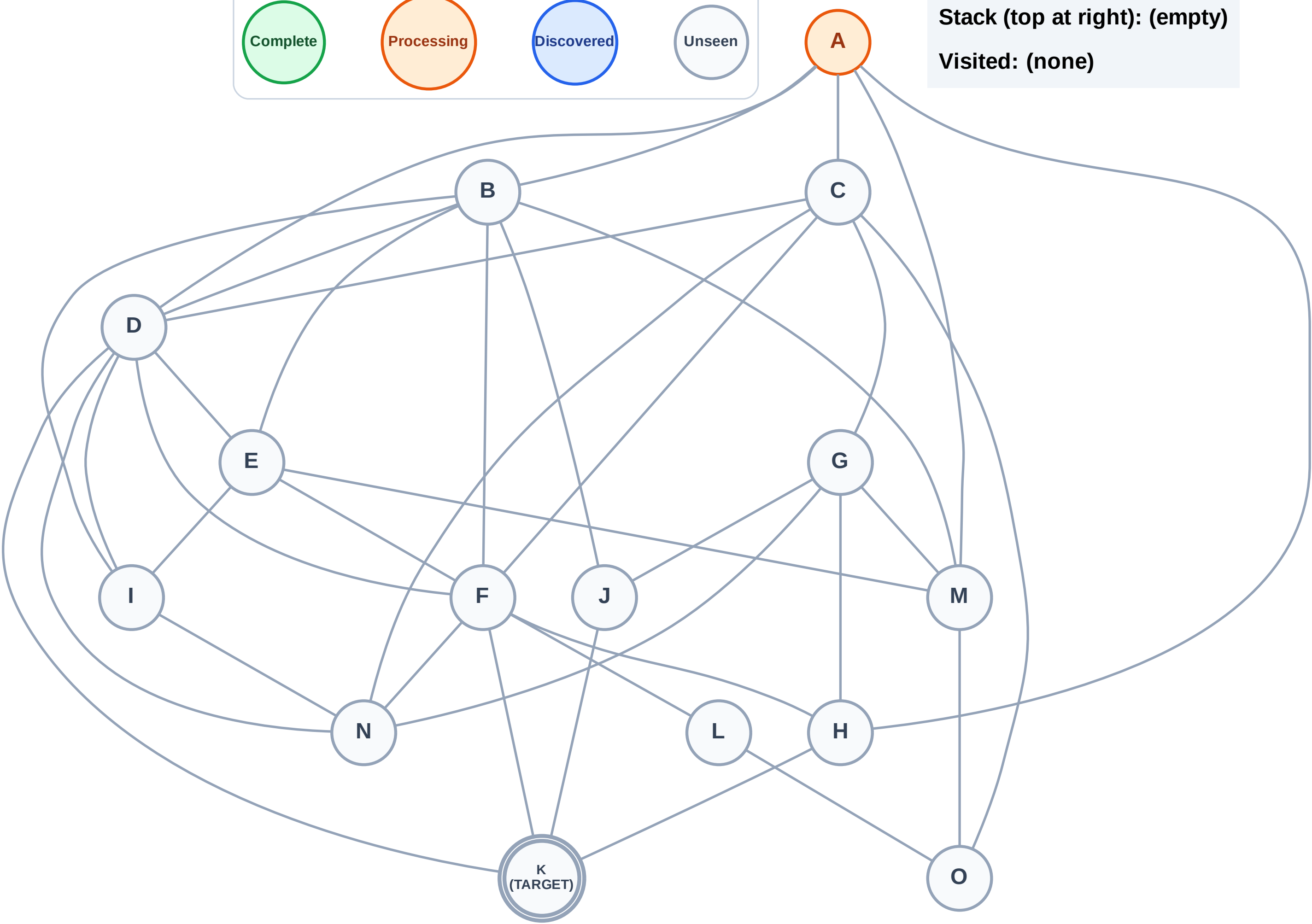
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



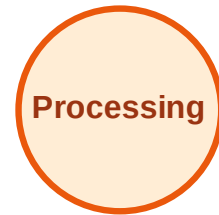
DFS Traversal

Step 3: Discover M, H, D, C, B from A

Legend



Complete



Processing



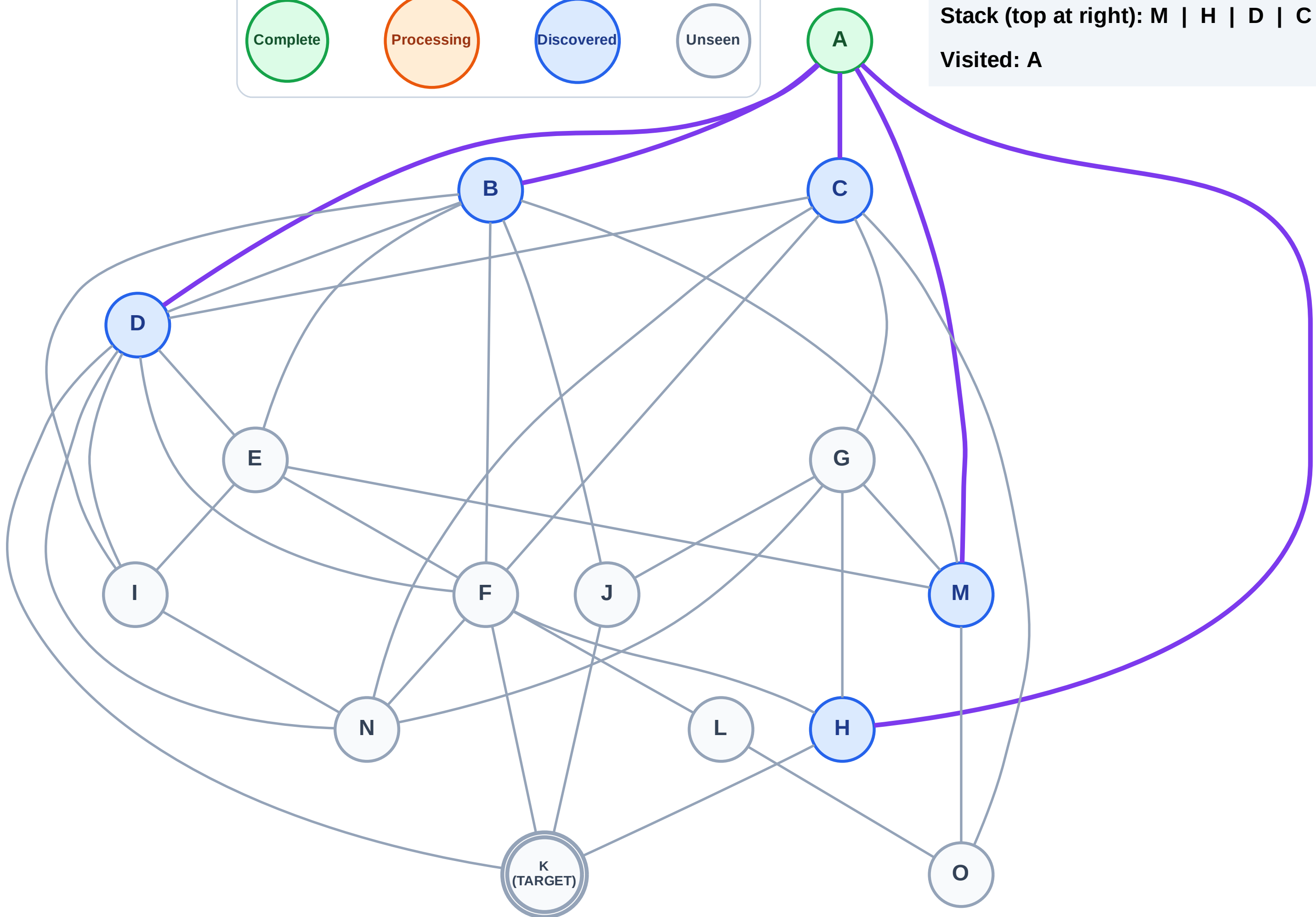
Discovered



Unseen

Stack (top at right): M | H | D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

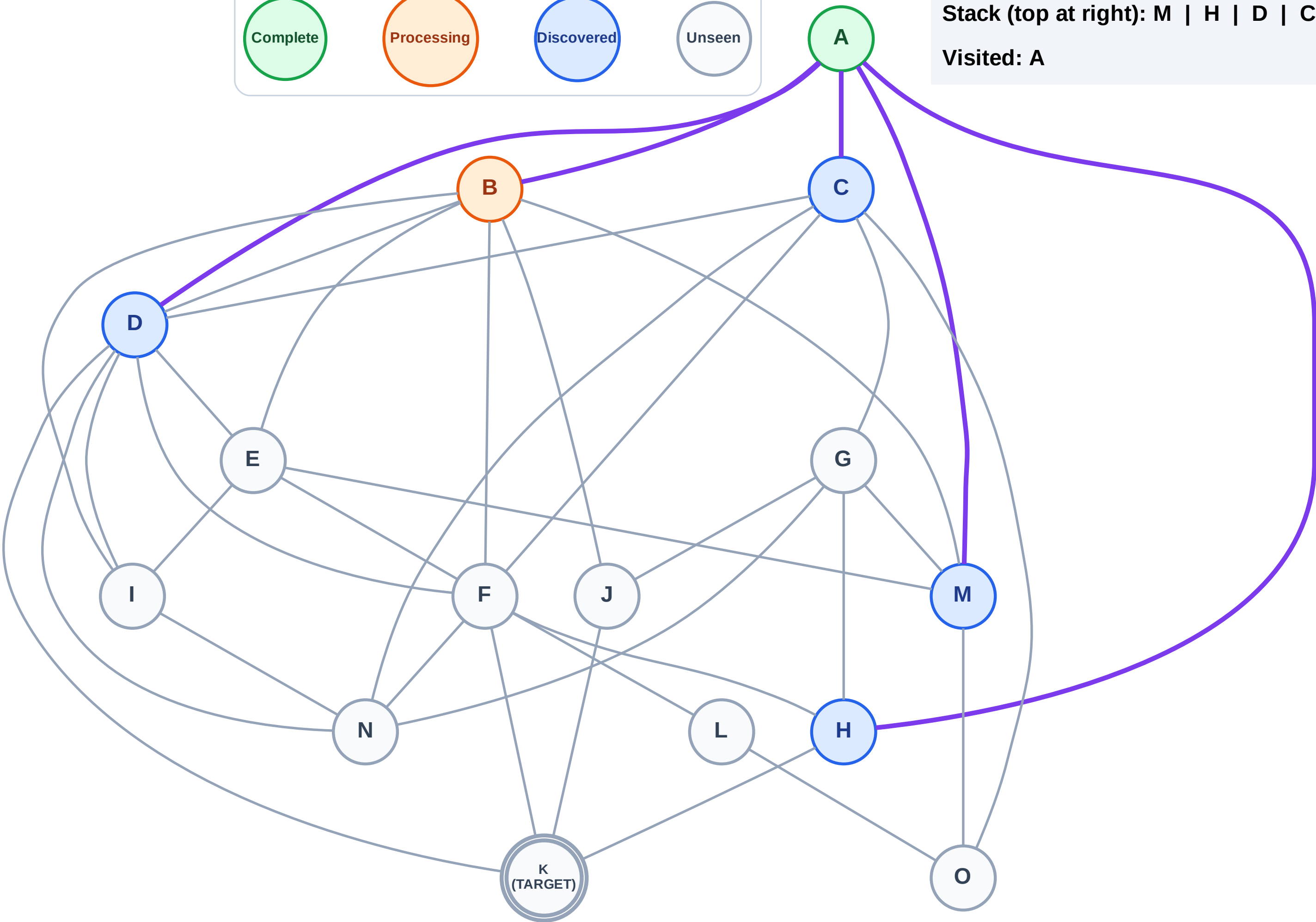
Processing

Discovered

Unseen

Stack (top at right): M | H | D | C

Visited: A



DFS Traversal

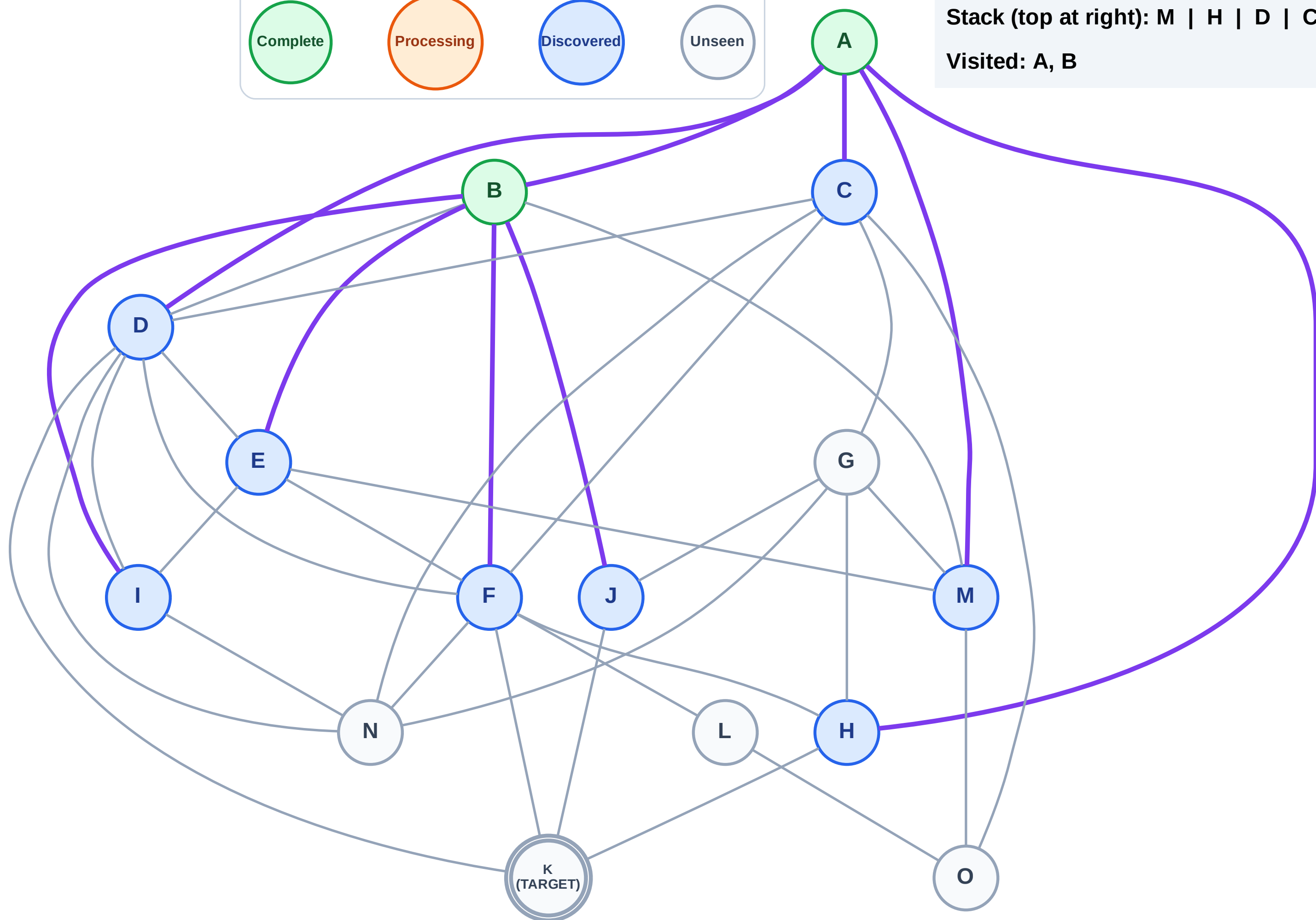
Step 5: Discover J, I, F, E from B

Legend



Stack (top at right): M | H | D | C | J | I | F | E

Visited: A, B



DFS Traversal

Step 6: Process node E

Legend

Complete

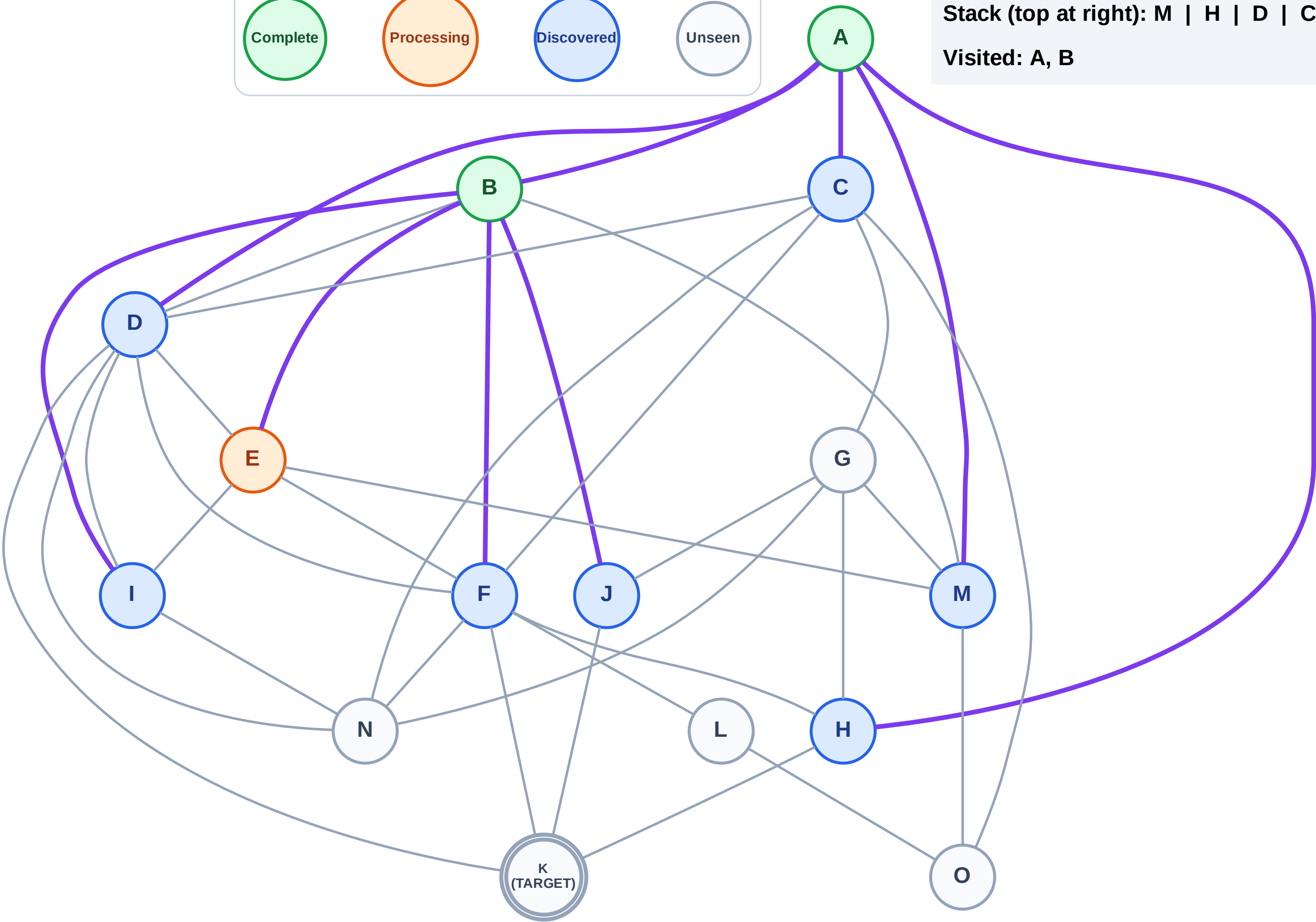
Processing

Discovered

Unseen

Stack (top at right): M | H | D | C | J | I | F

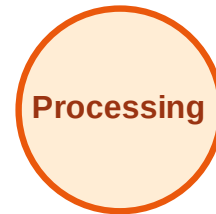
Visited: A, B



DFS Traversal

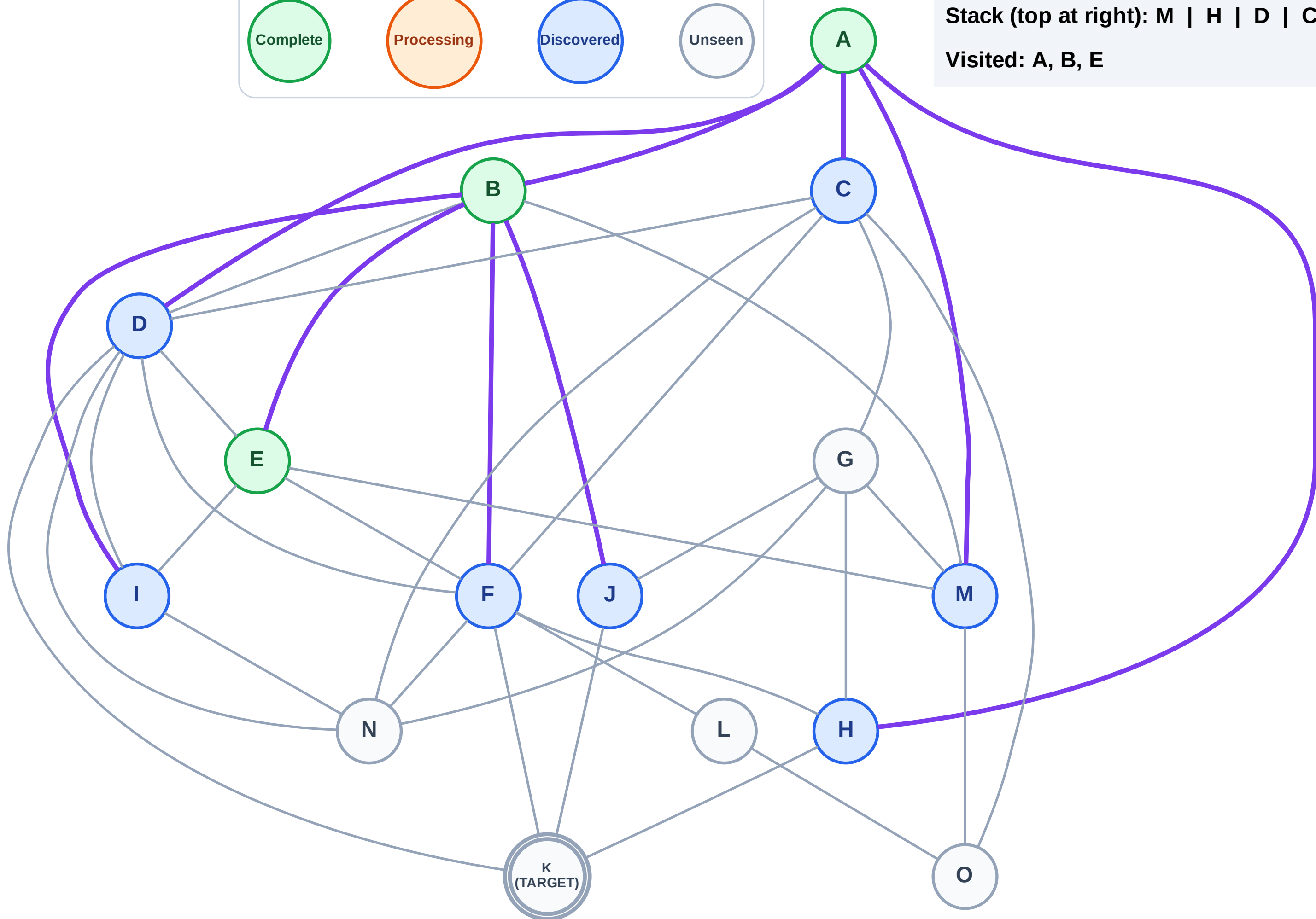
Step 7: No new neighbors from E

Legend



Stack (top at right): M | H | D | C | J | I | F

Visited: A, B, E



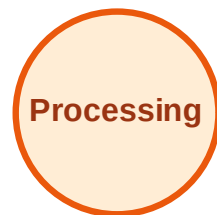
DFS Traversal

Step 8: Process node F

Legend



Complete



Processing



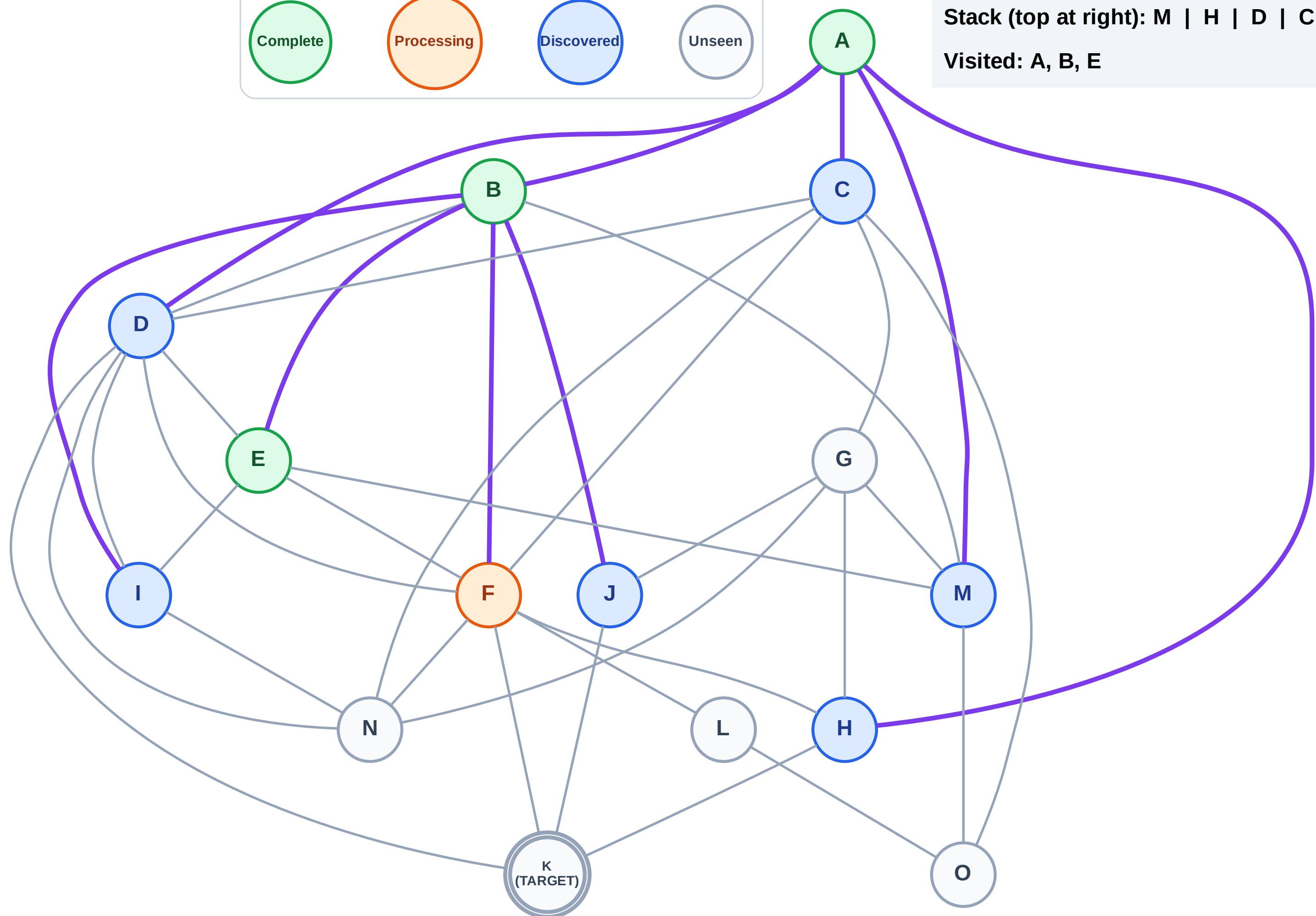
Discovered



Unseen

Stack (top at right): M | H | D | C | J | I

Visited: A, B, E



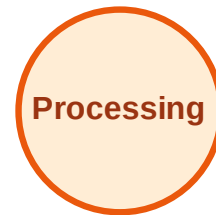
DFS Traversal

Step 9: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → F → K

Legend



Complete



Processing



Discovered



Unseen

Stack (top at right): M | H | D | C | J | I | N | L

Visited: A, B, E, F, K

